Predicting Neonatal Sepsis

with Machine Learning

Research & Data Science · Neotree

The EONS Algorithm

From Routine Clinical Data to Clinical Decision Support

1 · Defining the Problem

BACKGROUND

EONS (Early Onset Neonatal Sepsis) is a life-threatening infection in newborns. It is one of the leading cause of neonatal mortality in sub-Saharan Africa — yet it is treatable if caught early.

What is the Problem?

1. Difficult to diagnose without Blood Culture
2. Blood Cultures service is unreliable

Clinicians must decide whether to give antibiotics without rapid diagnostics. They rely on non-specific, overlapping clinical signs. This leads to Overprescrition of Antibiotics inturn increase in AMR

What are we Predicting?

Whether a neonate admitted to NICU has EONS, using only admission features recorded before a diagnosis is made.
Then recommend to give Antibiotics, wait or Not

Classification or Regression?

Binary Classification —
Sepsis (1) vs. No Sepsis (0).

Output: Still figuring this out. (Traffic light system, probability, yes or nor)

What does Success Look Like?

WHO/UN recommends ≥85% Sensitivity AND ≥85% Specificity. We must not miss cases, but also avoid over-treating.

Reduce the number of babies given antibiotics at the point of admission without an increase in negative outcomes due to EONS

2 · What is Neotree? - Source of data

A Digital Health Platform - Used for decision making support in Newborn Care

in Low-Resource Settings

- Deployed in neonatal units across Zimbabwe and Malawi
- Captures structured clinical data at admission and discharge
- Supports bedside decision-making for clinical staff
- Generates research-grade datasets for quality improvement projects

~50k+ Admissions & Discharges Records

+700 Features Total

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4 Central & Provincial hospitals in Zimbabwe

3 Central & Provincial Hospitals in Malawi

Neotree Data

neotree

THRU ZIM
THE HEALTH RESEARCH UNIT ZIMBABWE

ADMISSIONS

Emergency Triage
Vital Signs
Examinations
Plan

DISCHARGES

Discharge Diagnosis
Neotree Outcomes
Meds Given
Vital Signs

NEOLAB

BLOOD CULTURES
Organisms
Resistances

Admissions
Discharges
Neolab

ML Model

Algorithm

Complete Admission Script

Give ABX

Wait & Observe

Don’t Give ABX

3 · Cleaning & Preprocessing - Ready for Machine Learning

1 Missing Value Imputation

Imputation for continuous variables

Imputation for categorical variables

Separate 'missing' indicator flag per feature

2 Numeric Normalization

Min-Max scaling for bounded clinical ranges

Standard scaling (z-score) for unbounded variables

Fit scalers on training set only — no leakage

Log-transform right-skewed distributions

3 Categorical Encoding

One-hot encoding for nominal variables

Ordinal encoding for ordered clinical scores

Rare category grouping (< 2% frequency)

4 Feature Engineering

Clinical risk score combinations

Interaction terms (gestation × birth weight)

Binary flags for clinically relevant thresholds

4 · Model Selection

We explored four families of models:

Baseline

Linear Models

Logistic Regression, Lasso

Pros:

Interpretable, fast, good baseline

Trade-offs:

Cannot capture non-linear interactions

★ Best Fit

Tree-Based Models

XGBoost, Random Forest, LightGBM

Pros:

Handles missing values, non-linear patterns, feature importance, robust to scale

Trade-offs:

Risk of overfitting without tuning

Explore Later

Neural Networks

MLP, TabNet

Pros:

High capacity, can learn complex patterns

Trade-offs:

Needs more data, less interpretable, harder to tune

Experimental

Semi-Supervised ML

Label Propagation, Self-training

Pros:

Uses unlabelled records to improve decision boundaries

Trade-offs:

Sensitive to noisy pseudo-labels; ground truth issues amplified

Modelling Decisions

T0 CONS

sepsis or contamination?

Default - CONS = Sepsis +
Trial - CONS = Sepsis -

T1 CV Strategy

Default - Expanding Window

Trial - Sliding Window
Trial - Stratified CV

T2 Feature Space

Default Pre-selected

Trial - All features
Trial - Feature Selection OFF

T3 Ambiguous Cases

Default - Drop

Trial - Label as True
Trial - Label as False

T4 Cohort Filters

Default: Exclude Gest<32, Default: Exclude Age>72hrs

Gestation
age at admission

T5 Test Window held-out test

Default - Last 3 months

Trial -Last 1 month
Trial - Last 2 Months
★ HIGH IMPACT

T6 Data Cutoff

Default - Sept 2025

Trial -Dec 2025
Trial - Feb 2026
⚠ CRITICAL

T7 Preprocessing

Missingness
Imputation
indicators
encoding
resampling

SENS: 79, SPEC: 61

10 · Next Steps & Shadow Testing

Shadow Testing:

- Run the model silently alongside normal clinical care.
- Clinicians make decisions independently — model outputs are logged but not shown.
- We compare predictions with actual outcomes to measure real-world performance without influencing care.

Challenge 1- Defining Sepsis

Our Target: Did the baby have EONS?

This is not a simple "Yes/No" question in the data.

- Blood cultures, our best source of truth, are scarce.
- Clinical diagnoses can be non-specific (e.g., just "Sepsis").

Solution: We created a **Composite Outcome Label** using a clear hierarchy.

How we classify an EONS case:

1. **Blood Culture (Source of Truth):**
   - Positive blood culture within 48 hours? **It's EONS.** Period.
2. **Explicit Clinical Diagnosis:**
   - Discharge diagnosis says "EONS"? **It's EONS.**
   - Discharge diagnosis says "LONS"? **It's NOT EONS.**
3. **Ambiguous Diagnosis ("Sepsis"):**
   - We apply careful filters (e.g., length of stay, mortality).

Challenge 2 - Missingness

Wide & Sparse Data

- Our dataset has over 100 features
- **Many columns have over 50% missing values.**

Some Solutions
Missingness indicators
Use Tree Based Models
Impute Categoricals Only

Challenge 3 - Class Imbalance

Sepsis is a **rare event** in the dataset.

Our Class Distribution:

- **90%** of babies do **NOT** have EONS.
- **10%** of babies **DO** have EONS.

Some Solutions:

Resampling

Challenge 4 - DATA DRIFT

A significant shift was detected in data from September 2025 onward

- possibly due to form changes,
- protocol changes
- Earlier-trained models degrade severely.

Some Solutions
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